

Tool Operator Guide

Google Cloud + OAuth Setup for SEO & GEO Auditor

Audience: Tool Operator

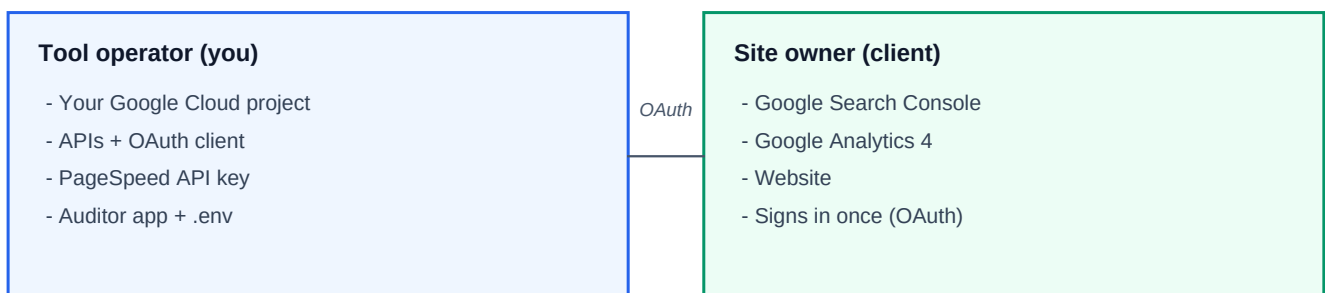
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Google Cloud + OAuth Setup for SEO & GEO Auditor

This guide is for the tool operator - the person who installs and runs the SEO & GEO Auditor app on a server or laptop.

You create one Google Cloud project (yours, not the site owner's). That project holds the APIs, OAuth app, and PageSpeed API key. The site owner never shares their Cloud project; they only sign in with their Google account so the app can read their Search Console and GA4.

Who owns what?



- You do not get access to the site owner's Google Cloud - they don't need one.
- They do not get access to your Google Cloud - they only approve OAuth read access to GSC/GA4.
- One your Cloud project can connect many client sites (each row in `connected_properties.json`).

Part 1 - Create your Google Cloud project

1. Sign in to Google Cloud Console with your Google account (operator account).
2. Project dropdown -> New project.
3. Name it (e.g. SEO GEO Auditor).

4. Click Create. Note the Project ID (e.g. seo-geo-auditor-123456).

Enable billing only if Google prompts you. Typical audit usage stays within free tiers.

Part 2 - Enable APIs

APIs & Services -> Library - enable each:

API	Purpose
Google Search Console API	Sitemap status, CTR, URL Inspection
Google Analytics Data API	GA4 engagement metrics
Google Analytics Admin API	List GA4 properties during setup
PageSpeed Insights API	Core Web Vitals in audits

Part 3 - PageSpeed API key

OAuth does not cover PageSpeed. Create a separate API key:

1. APIs & Services -> Credentials -> + Create credentials -> API key.
2. Copy the key into the project root .env:

```
GOOGLE_API_KEY=your_api_key_here
```

3. (Recommended) Restrict the key to PageSpeed Insights API only.

Restart the backend after editing .env.

Part 4 - OAuth consent screen

1. APIs & Services -> OAuth consent screen.
2. User type: External (unless internal Workspace only).
3. App name, support email, developer email -> Save and continue.
4. Scopes -> Add or remove scopes:
 - <https://www.googleapis.com/auth/webmasters.readonly>
 - <https://www.googleapis.com/auth/analytics.readonly>
5. If app status is Testing, add Test users: email(s) that will sign in (site owner's Gmail, or yours if you own the properties).

Part 5 - OAuth Desktop client

1. Credentials -> + Create credentials -> OAuth client ID.
2. Type: Desktop app.
3. Download JSON -> save as:

```
secrets/google-oauth-client.json
```

Never commit this file (it is git-ignored).

Part 6 - Browser sign-in

From the backend folder:

```
cd backend
source .venv/bin/activate
pip install -r requirements.txt
python scripts/google_auth.py
```

- Browser opens -> sign in with the account that has access to the client's GSC and GA4 (often the site owner's account; you can sit with them or share screen).
- Token saved to secrets/google-oauth-token.json.

Re-list properties without browser:

```
python scripts/google_auth.py --list-only
```

You should see GSC URLs and GA4 Property IDs. Compare with what the site owner sent you.

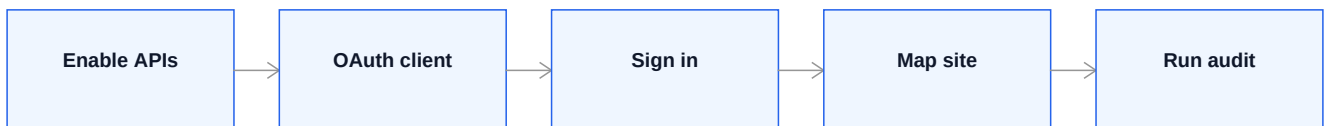
Part 7 - Register each site

Edit backend/connected_properties.json:

```
{
  "example.com": {
    "gsc_site_url": "https://www.example.com/",
    "ga4_property_id": "123456789"
  }
}
```

Field	Rule
Key	Hostname only, lowercase, no www
gsc_site_url	Exact match from GSC / --list-only
ga4_property_id	Numeric only

Restart the backend. Run a test audit; confirm connected and first-party GSC rows are scored.



Part 8 - Verify connection

Check	Expected
Audit shows connected	Yes for mapped domain
Sitemap GSC-submission	Scored (Meeting / Partial / Not Meeting)
Log file analysis	Scored when URL Inspection works
CTR	Scored; Not Meeting if 0 impressions is OK
GA4 engagement (3 rows)	Manual until GA4 has sessions

Part 9 - Troubleshooting

Issue	Fix
--list-only shows no GSC sites	Wrong Google account; use site owner's account
GSC Not Measured	Check gsc_site_url exact match; Search Console API enabled; retry audit
OAuth blocked (Testing)	Add user email under OAuth consent -> Test users
PageSpeed Not Measured	Enable PageSpeed API; set GOOGLE_API_KEY

Issue	Fix
GA4 list empty	Enable Analytics Admin API; or get Property ID from site owner manually

Tool operator checklist

- ☐ Google Cloud project created (your account)
- ☐ Four APIs enabled
- ☐ PageSpeed API key in .env
- ☐ OAuth consent screen + scopes configured
- ☐ Desktop OAuth client -> secrets/google-oauth-client.json
- ☐ python scripts/google_auth.py completed (site owner account if needed)
- ☐ connected_properties.json updated per client site
- ☐ Backend restarted; test audit successful

Files reference

File	Purpose
secrets/google-oauth-client.json	OAuth Desktop client (you create)
secrets/google-oauth-token.json	Token after sign-in (auto-created)
backend/connected_properties.json	Domain -> GSC URL + GA4 ID
.env -> GOOGLE_API_KEY	PageSpeed only

Tool Operator Guide - SEO & GEO Auditor. Give Connected_Mode_Site_Owner_Guide.pdf to your client for GSC, GA4, and OAuth sign-in steps.